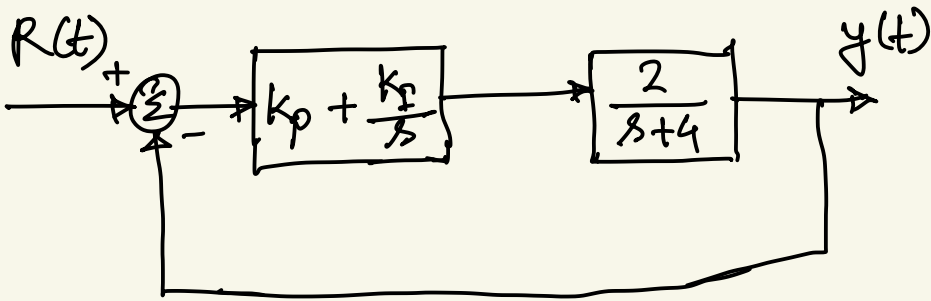




Example problem:

Design a PI controller for a control system so the system will have damping $\xi = 1$ and natural frequency $\omega_n = 4$ rad/sec



$$D_c(s) = k_p + \frac{k_I}{s} = \frac{k_p s + k_I}{s}$$

$$G(s) = \frac{2}{s+4}$$

$$T_{cl}(s) = \frac{D_{cl}(s) G(s)}{1 + D_{cl}(s) G(s)}$$

$$= \frac{2k_p s + 2k_I}{s(s+4) + 2k_p s + 2k_I}$$

We find the system characteristic equations when the denominator is set to zero.

$$s^2 + (4 + 2k_p)s + 2k_I = 0 \quad \text{--- ①}$$

Desired characteristic equation

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\Rightarrow s^2 + 2(1)(4)s + (4)^2 = 0$$

$$\Rightarrow s^2 + 8s + 16 = 0 \quad \text{--- (1)}$$

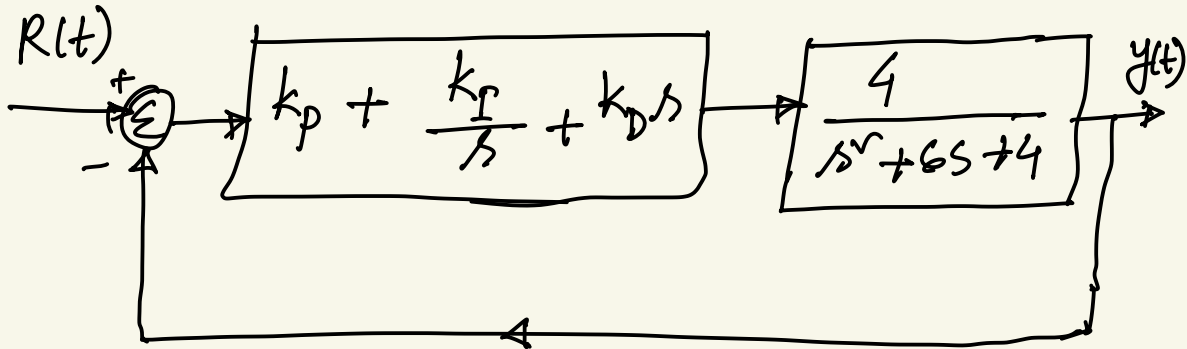
By comparing eq (1) and (2)
we get,

$$\left. \begin{array}{l} 4 + 2k_p = 8 \\ 2k_I = 16 \end{array} \right\} \text{solving} \quad \begin{array}{l} k_p = 2 \\ k_I = 8 \end{array}$$

Design a PID controller for the control system shown in the below figure that satisfies the following specifications

Natural frequency $\omega_n = 2$
damping ratio $\zeta = 0.85$

and have a real pole at $s = -10$ in its characteristic equation.



$$\begin{aligned} D_{cl}(s) &= k_p + \frac{k_I}{s} + k_D s \\ &= \frac{k_p s + k_I + k_D s^2}{s} \end{aligned}$$

$$G(s) = \frac{4}{s^2 + 6s + 4}$$

$$T_{cl}(s) = \frac{D_{cl}(s) G(s)}{1 + D_{cl}(s) G(s)}$$

$$\begin{aligned}
 T_u(s) &= \frac{(k_p s + k_I + k_D s^2)}{s} \cdot \frac{4}{s^2 + 6s + 4} \\
 &= \frac{1 + \frac{k_p s + k_I + k_D s^2}{s}}{1 + \frac{k_p s + k_I + k_D s^2}{s}} \cdot \frac{4}{s^2 + 6s + 4} \\
 &= \frac{(4k_p s + 4k_I + 4k_D s^2) / s (s^2 + 6s + 4)}{(s^3 + 6s^2 + 4s + 4k_p s + 4k_I + 4k_D s^2)} \\
 &= \frac{4k_D s^2 + 4k_p s + 4k_I}{s^3 + (6 + 4k_D)s^2 + (4 + 4k_p)s + 4k_I}
 \end{aligned}$$

Compare with the desired characteristic equation with non-dominant poles,

$$(s+10)(s^2 + 2\zeta\omega_n s + \omega_n^2) = 0$$

$$s^3 + (10 + 2\zeta\omega_n)s^2 + (20\zeta\omega_n + \omega_n^2)s + 10\omega_n^2 = 0$$

$$10\omega_n^2 = 4k_I, \quad \omega_n = 2$$

$$\Rightarrow k_I = 10$$

$$4k_p + 4 = 20\zeta\omega_n + \omega_n^2$$

$$4k_p + 4 = 20(0.85)(2) + (2)^2$$

$$\Rightarrow 4k_p + 4 = 38$$

$$\Rightarrow k_p = 8.5$$

$$\begin{aligned} 4k_D + 6 &= 10 + 2\zeta\omega_n \\ &= 10 + 2(0.85)(2) = 13.4 \end{aligned}$$

$$k_D = 1.85$$

$$D_c(s) = \frac{1.85s^2 + 8.5s + 10}{s}$$

